GLOBAL SECTIONS COMMUTE WITH DERIVED LIMITS, WITH THE ASSOCIATED SHORT EXACT SEQUENCES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for the commutation statement and for the displayed short exact sequences.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Derived limit of an inverse system of objects of the derived category. Same chapter together with the chapter Derived Categories, the defining triangle of R lim and its expression through products and shifts. Supplies the object whose sections are computed.
- Compatibility of the section functors with products and with the derived pushforward along the morphism to the point topos. Same chapter, the identification of global sections and of sections over an object with derived pushforwards along the corresponding morphism of topoi, together with commutation with products. Supplies the argument showing the commutation and the short exact sequences.

Source reference. Original lemma 08U1: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 5142–5142. Navigation pointer invoked at proof line 5142; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 08U5: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 5144–5144. Navigation pointer invoked at proof line 5144; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

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Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. The functors $R\Gamma(\mathcal{C}, -)$ and $R\Gamma(U, -)$ for $U \in \mathrm{Ob}(\mathcal{C})$ commute with $R\lim$. Moreover, there are short exact sequences*

$$0 \to R^1 \lim H^{m-1}(U, K_n) \to H^m(U, R\lim K_n) \to \lim H^m(U, K_n) \to 0$$

for any inverse system (K_n) in $D(\mathcal{O})$ and $m \in \mathbf{Z}$. Similar for $H^m(\mathcal{C}, R\lim K_n)$.

Proof. The first statement follows from Injectives, Lemma 08U1. Then we may apply More on Algebra, Remark 08U5 to $R\lim R\Gamma(U, K_n) = R\Gamma(U, R\lim K_n)$ to get the short exact sequences. □